The Centered Door

5 instances · Download all inputs (.zip)

The Centered Door

The old research wing is sealed by two mechanical doors, each governed by a network of **chips** connected by **wires**. A door only reacts when its network is **connected**, i.e. when it is a single piece, with a path of wires between every pair of chips. When it is, the mechanism powers up the one chip that sits at the network's balance point, the **central chip**, and the door swings open.

> **Central chip.** In a connected network of N chips, a chip is *central* if removing it (together with its wires) leaves pieces that each contains **at most ceil(N/2)** chips.

You are standing in front of the **exit door**. Its network is a perfectly good **tree**, but the wire that should carry current from its central chip to the release button has been cut, and you have no spare wire.

So you turn to the **entrance door** behind you and harvest wire from it. You may pull out a wire **only if the entrance network stays connected afterwards**, and if **every chip that was central remains central**. Every wire you remove is spliced into one long cable, whose length is the **sum of the lengths** of the removed wires.

Each chip of the exit door sits at a known **distance** from the release button. Your cable can power an exit chip if it is **at least as long** as that chip's distance. If your cable reaches a **central** exit chip, the door opens.

Goal

Determine the index of the central exit chip that your cable can reach (the chip that opens the door) or report that no central exit chip is within reach. At most one such chip exists in any instance.

Input Format

The first line contains two integers `Ne Me` : the number of entrance chips (labeled `0 .. Ne-1`) and the number of entrance wires.

The next `Me` lines each contain three integers `A B L`, a wire of length `L` between entrance chips `A` and `B`.

The following line contains two integers `Ns Ms` : the number of exit chips (labeled `0 .. Ns-1`) and the number of exit wires.

The next `Ms` lines each contain two integers `A B`, a wire between exit chips `A` and `B`.

The last line contains `Ns` integers `D0 D1 ... D(Ns-1)`, where `Di` is the distance of exit chip `i` from the release button.

Output Format

A single integer: the label of the central exit chip that the cable opens, or `-1` if no central exit chip is reachable.

Constraints

Instances are guaranteed to satisfy the following constraints:

- `2 ≤ Ne ≤ 100000` and `1 ≤ Me ≤ 200000`
- `2 ≤ Ns ≤ 100000` and the exit network is a **tree** (`Ms = Ns - 1`)
- `0 ≤ Di ≤ 10^9` and `0 ≤ L ≤ 100000`
- The entrance network is **connected**.
- Each network has **at least one** central chip.
- There is **never more than one** valid answer.

Example

Instance in input:

```
7 9
0 1 5
0 2 8
```

```
0 3 6
1 2 7
2 3 4
2 4 5
2 5 2
4 5 4
4 6 9
6 5
0 2
1 2
2 3
3 4
4 5
18 28 20 32 24 15
```

Solution in output:

```
2
```

Instances

Submit all instances at once

Upload one .zip containing a file per instance, each named after its instance number (e.g. `instance.1.out`, `2.txt`).

Browse... No file selected.

Submit all

Your code

Upload a .zip of the code your team used to generate the answers for this problem. Re-uploading replaces the previous file.

Current upload: download your code

Browse... No file selected.

Upload code

Instance 1 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 1

Instance 2 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 2

Instance 3 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 3

Instance 4 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 4

Instance 5 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 5